WEEKLY PROGRESS UPDATE #4

Emotion-Mediated Personalized Image Aesthetic Assessment

Predicting individual aesthetic preferences through emotional responses

Presenter: Pinwa

02 — RECAP

How the system predicts a score.

Direct photo → score (skips the emotion step)

Hybrid photo → 7 emotions → score

Image
INPUT
6,526 photos
art / fashion / landscape

→

Backbone
FROZEN
image → numbers
not trained by us

→

Guess 7 emotions
STEP 1
how does this photo feel?
impressed · intellectual · motivated · amused · nostalgic · sad · distasteful
trained on 104 other users

→

Personal formula
STEP 2
7 weights, one per emotion
trained on this user's 100 photos

→

Score
OUTPUT
0 – 6
how much THIS user likes it

Two separate steps, each fitted on different people

The emotion step learns from **104 other users**, so it is the same for everyone. The formula is fitted on the **target user's own 100 photos** — that is where personalisation happens. Both steps are simple ridge regression, not neural networks.

Terms & data

CCC — how close predictions are to real scores (0–1). Punishes wrong order *and* wrong size.
emo_r — how well we guess this person's emotions. Average now **0.27**.
XPASS-Vis — 129 users · 6,526 photos · 3 domains · 87,836 ratings · scores 0–6

03 — RECAP & AGENDA

Previous Finding & Today's Goals.

21 July 2026

LAST TIME (UPDATE #3)

Baseline Performance & Qwen Findings

Qwen3-VL Finding (Ryu & Yanaka)

Tested 8B: The stronger the backbone, the smaller the Hybrid–Direct gap.

Strongest Overall Baseline

Fine-tuned CLIP remains our highest performing model (CCC ≈ 0.400).

TODAY (UPDATE #4)

01 Noise Ceiling Analysis
Uncovering the true upper bound by accounting for human self-agreement.

02 When emotion help
In which cases does emotion help users, and in which cases does it hurt?

03 Qwen3-VL-4B Model
Testing the smaller 4B model recommended in Ryu & Yanaka to check compute vs accuracy.

04 — PART 1 — THE CEILING

Is that ceiling actually real?

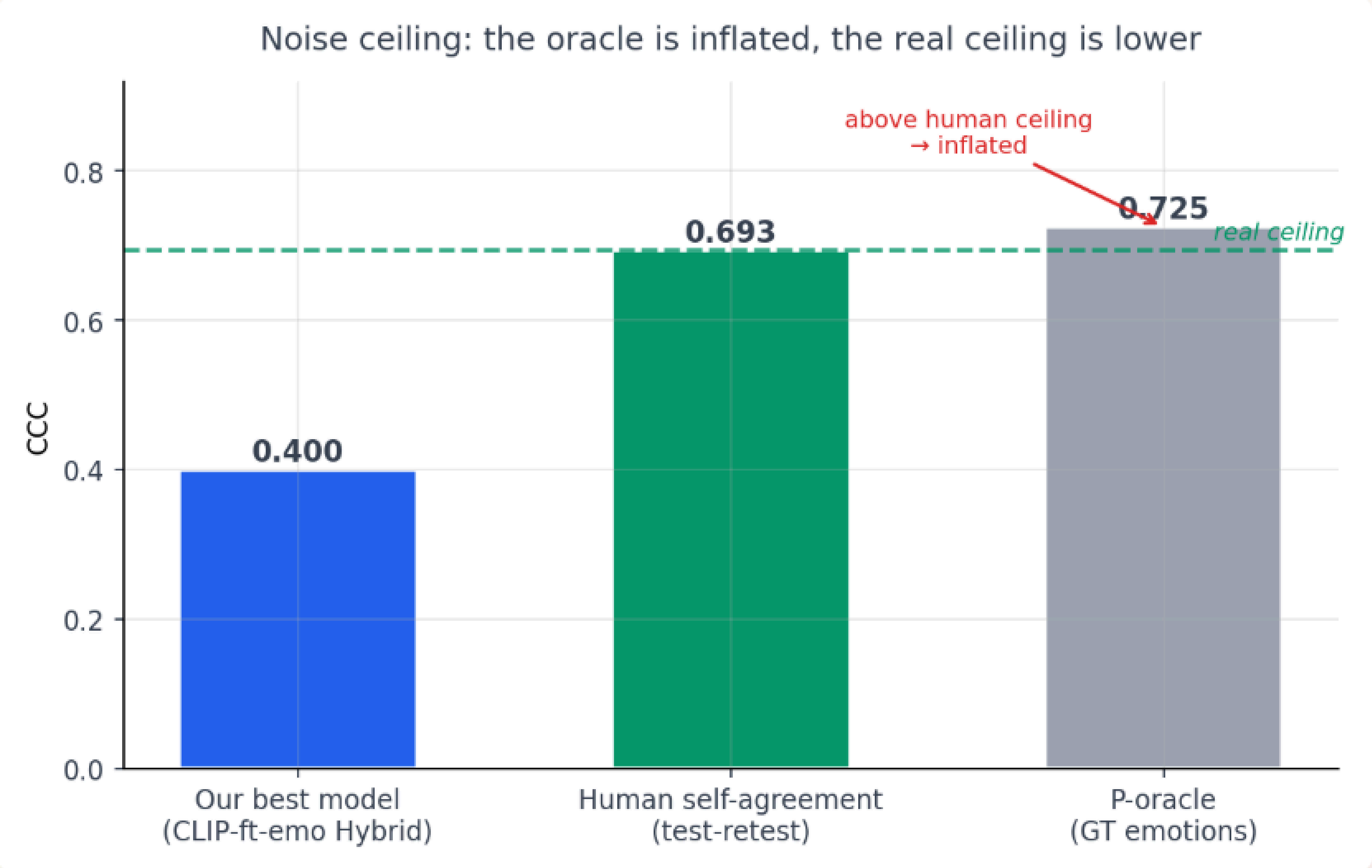
Our best model	Human self-agreement	P-oracle (inflated)
0.400	0.693	0.725

PEOPLE DO NOT AGREE WITH THEMSELVES

The dataset has a built-in test-retest design: users rated the **SAME image twice** under identical conditions. People gave the exact same score only **44%** of the time.

4,509 Repeated Pairs Analyzed

Honest Upper Bound: The P-oracle (0.725) sits **above** human self-agreement because emotions & scores share same-session state noise. The real ceiling is ~**0.69** — so our 0.400 reaches **~58%** of what is actually reachable.



P-oracle = model given ground-truth user emotion ratings · **test-retest** = same user, same image, rated twice

05 — PART 2 · WHEN DO EMOTIONS HELP

From an average to a reason.

USER LEVEL 129 users × 3 domains = 387 units

1 Which person does it help?

Checking if emotion helps all users equally. We run a paired Wilcoxon test to see if it is a broad, common effect or driven by a few outliers.

2 What decides whether it helps?

Analyzing key factors that determine whether emotion helps:

- A Emotion prediction accuracy **(emo_r)**
- B Baseline strength of the Direct model **(Direct)**
- C Rating consistency & category headroom

IMAGE LEVEL per-image error reduction

3 Which images does it help?

Checking if photos triggering different emotions gain different benefits. We group images by their dominant emotion (e.g. distasteful vs. intellectual) to compare.

06 — PART 2 — HOW WE MEASURE IT

The setup.

WHAT WE ARE EXPLAINING

Delta (delta)

```
delta = Hybrid CCC - Direct CCC
```

+ **positive delta**
Emotion helped

− **negative delta**
Emotion hurt

129 users × 3 domains = **387 units** to analyze

WHAT MIGHT EXPLAIN IT

How well we read feelings (emo_r)

Measures how accurately Stage 1 predicts **THIS specific user's** 7 emotions:

```
emo_r = mean correlation(predicted, actual) across 7 emotions
```

Average across all users
0.27
the weak link in the whole system

No new model here — I reuse the Direct and Hybrid scores each user already got in earlier runs, on the same images and the same fold, and just subtract them.

07 — PART 2 — FIRST ANSWER

It helps almost everyone.

WHAT THE HISTOGRAM SHOWS

Each bar is one user × domain unit. Bars to the **right of zero** = emotion helped. Bars to the **left** = emotion hurt.

91.7% of units sit right of zero

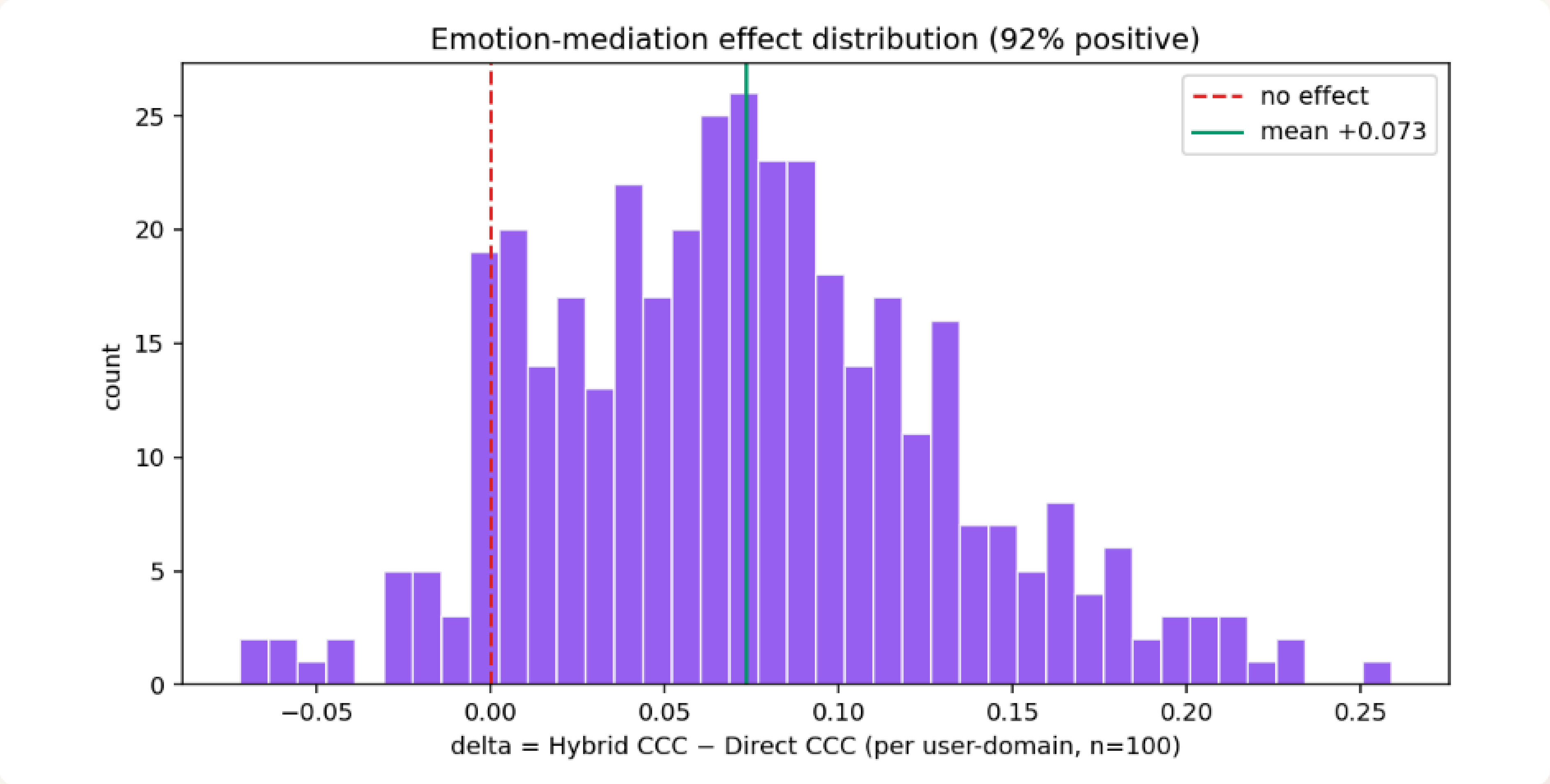
p-value	Median gain	Effect r
< 0.001	+0.071	0.82

THE LEFT TAIL (8.3%, 32 UNITS)

Not random. They cluster exactly where emotion prediction was poor:

Left-tail avg emo_r	vs	Rest avg emo_r
0.153	VS	0.283

→ So emotion accuracy (emo_r) seems to decide whether it helps or hurts. Is that true?



08 — MAIN FINDING

Who does it help, and why?

OUR GUESS BEFORE TESTING

Emotions should help more when we guess that person's feelings well — and hurt when we guess them badly.

Across all 387 users

$r = +0.30$ ($p = 1e\text{-}9$)

HARDEST TO READ (BOTTOM 25%)

emo_r ≈ 0.09

+0.040 gain

80% helped

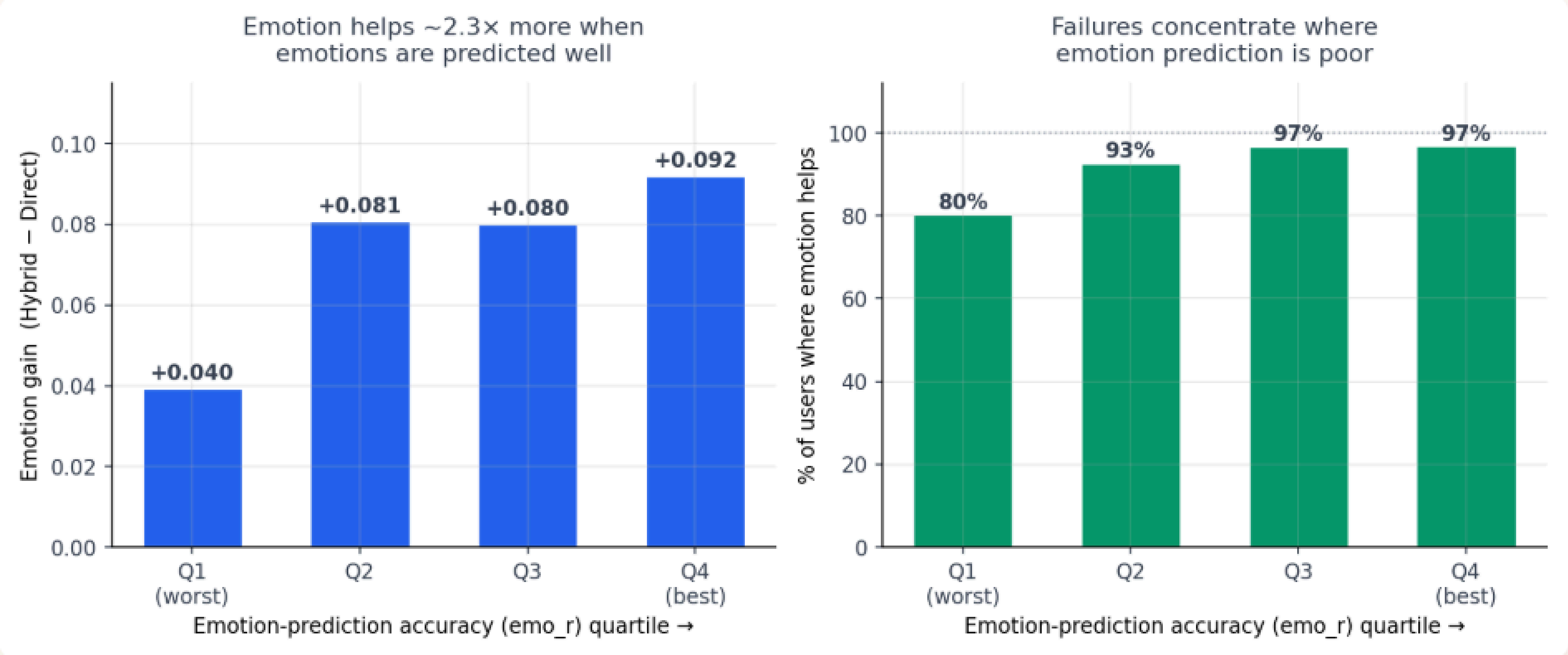
EASIEST TO READ (TOP 25%)

emo_r ≈ 0.45

+0.092 gain

97% helped

The 32 users it made worse: for them we read feelings at only **0.15**, against **0.28** for everyone else. Emotions hurt exactly where we read them badly.



emo_r = how well we guess this person's feelings (average match across the 7 emotions, 0–1)

09 — PART 2 — IS THERE A CUT-OFF?

How good must the emotion guess be?

Is there an accuracy cut-off below which emotion isn't worth it?

No Threshold in Our Data

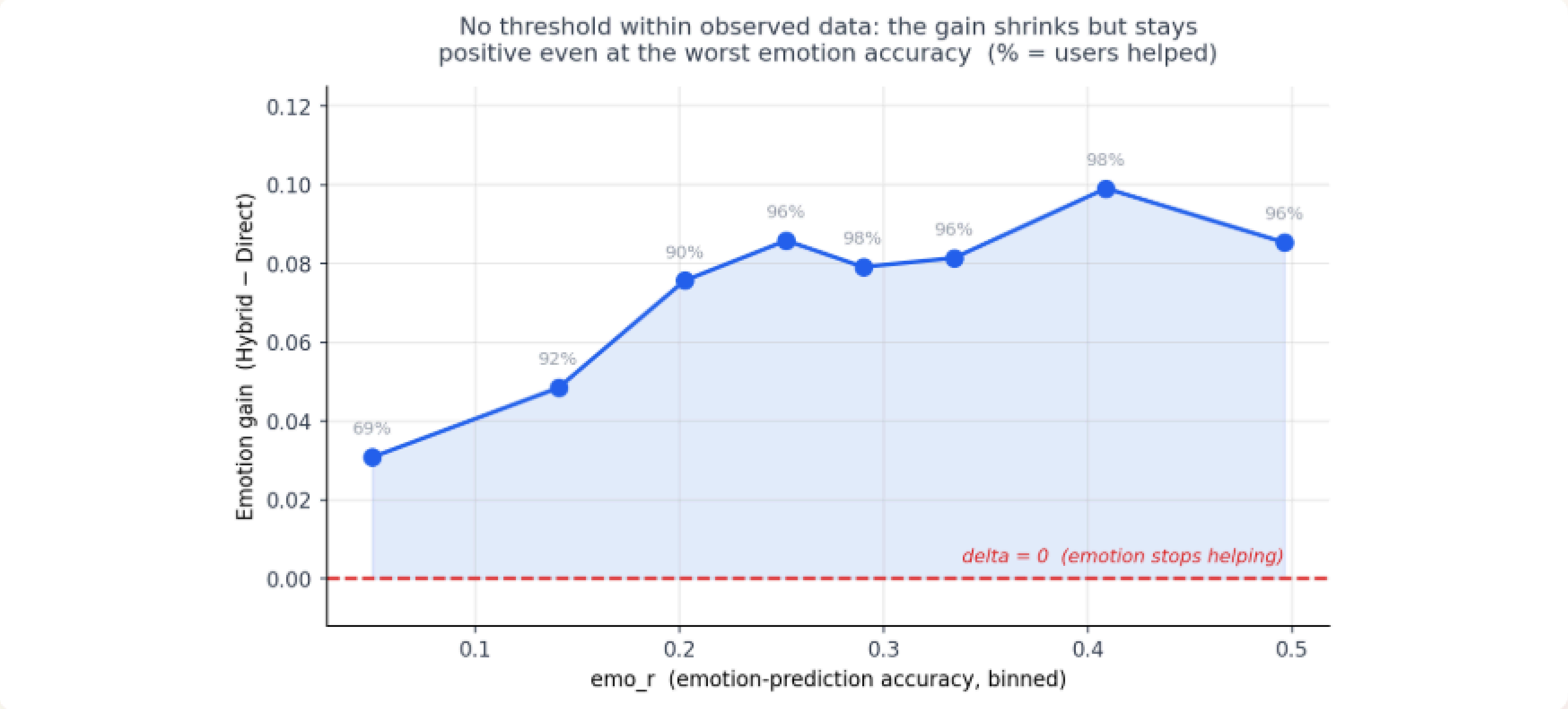
Even the group we read worst still gains +**0.031** on average. The fitted line only reaches zero at **emo_r** = −**0.29**, which is below anything we actually see in the data.

DOSE-RESPONSE MODEL

emo_r controls HOW MUCH it helps and HOW RISKY it is

Worst Quartile Risk	Best Quartile Risk
31% hurt	**3% hurt**

Conclusion: Higher emotion accuracy increases expected gain while minimizing risk of negative delta, but emotion mediation remains net positive across all observed emotion accuracy ranges.



In other words: reading feelings better does not switch the benefit on — it just makes it bigger, and safer.

10 — PART 2 — CHECKING THE FINDING

Only one thing really explains it.

WHAT IT LOOKS LIKE AT FIRST

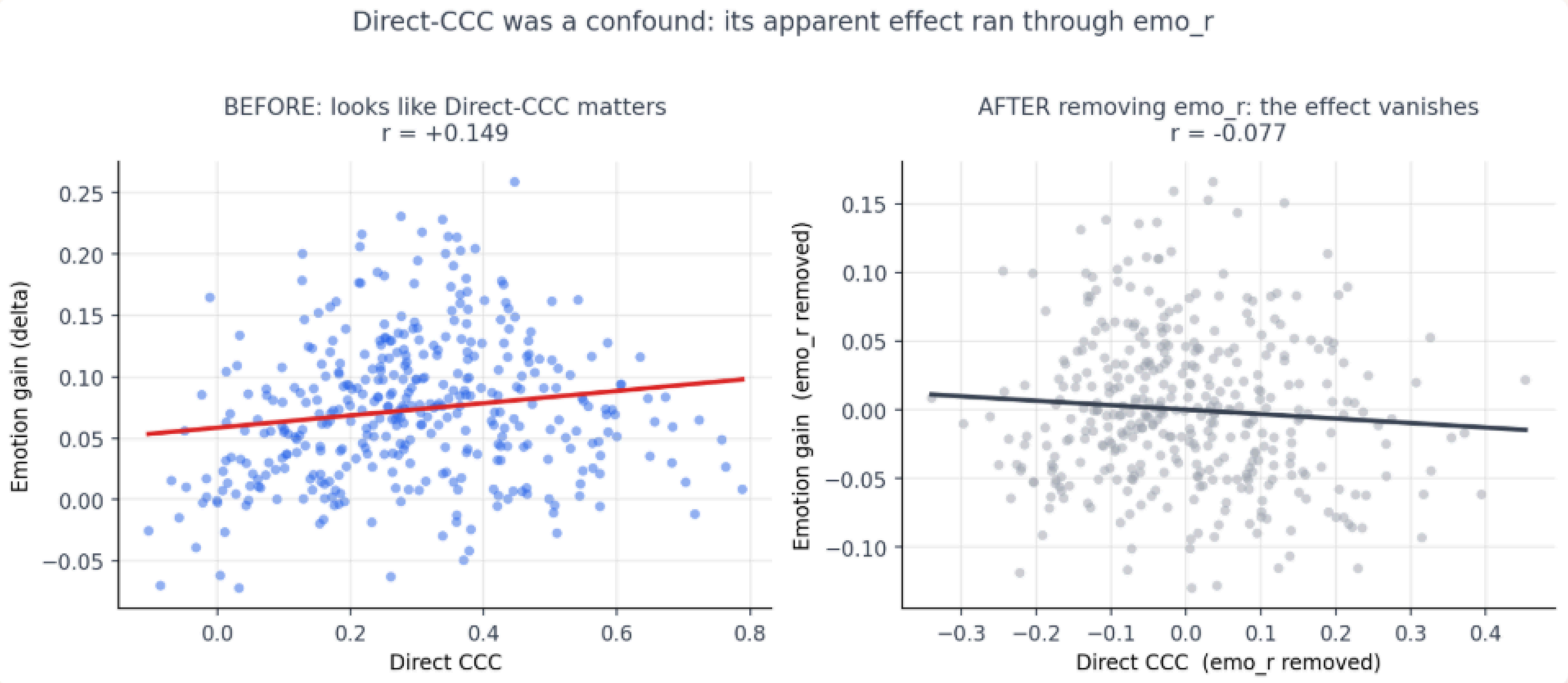
Users whose plain model is already good also seem to gain more from emotions (**r = +0.15**).

But those two go together: r = +0.66. If we read someone's photos well, we do well at both.

WHAT HAPPENS WHEN WE HOLD EMO_R FIXED

delta ↔ Direct CCC | controlling for emo_r	**−0.08 (gone)**
delta ↔ emo_r | controlling for Direct CCC	**+0.26 (SURVIVES)**

Single Mechanism: Direct strength was a statistical confound. There is only ONE underlying mechanism controlling emotion benefit: **emotion predictability (emo_r)**.



Analogy: Foot size correlates with reading ability in children — both grow with age. Here: age = **emo_r**, foot size = **Direct CCC**, reading = **delta**.

11 — PART 2 — THE ODD ONE OUT

Why is art different?

EMO_R → DELTA GAIN CORRELATION BY DOMAIN

Fashion	**r = +0.38 ✓ (significant)**
Landscape	**r = +0.34 ✓ (significant)**
Art	**r = +0.14 ✗ (not significant, p = 0.11)**

H1: CEILING EFFECT

Art Direct baseline is already high (**0.384** vs fashion **0.202**). Less headroom → delta varies less → correlation harder to detect statistically.

H2: DIFFERENT MECHANISM

Aesthetic judgments in artistic works stem from non-emotional features like technique, composition, or historical context rather than emotions.

Domain split: Fashion (129 users), Landscape (129 users), Art (129 users)

12 — PART 2 — THE ODD ONE OUT

Testing the two possible reasons.

Three tests — do they point to "ceiling" or "different mechanism"?

1. Partial correlation — control for Direct-model strength

If it's a ceiling, art's correlation should recover once baseline strength is removed

art: +0.13 (ns) → +0.18 · $p = 0.048$ — leans ceiling

2. Spread of the gain within each category (SD)

If art is capped, its gain should vary the least — lowest spread

art SD = 0.054 (lowest) · Levene $p = 0.21$ — weak / borderline

3. Interaction test — is the emo_r→gain slope different in art?

If art had a different mechanism, this would show up clearly

$p = 0.056$ (not significant) — no evidence for H2

All three lean toward "ceiling" — but every number sits on the 0.05 border

So: no evidence art works differently — not proof they're the same

13 — PART 2 — NOW BY PHOTO

Which photos benefit most?

GROUPED BY DOMINANT EVOKED EMOTION

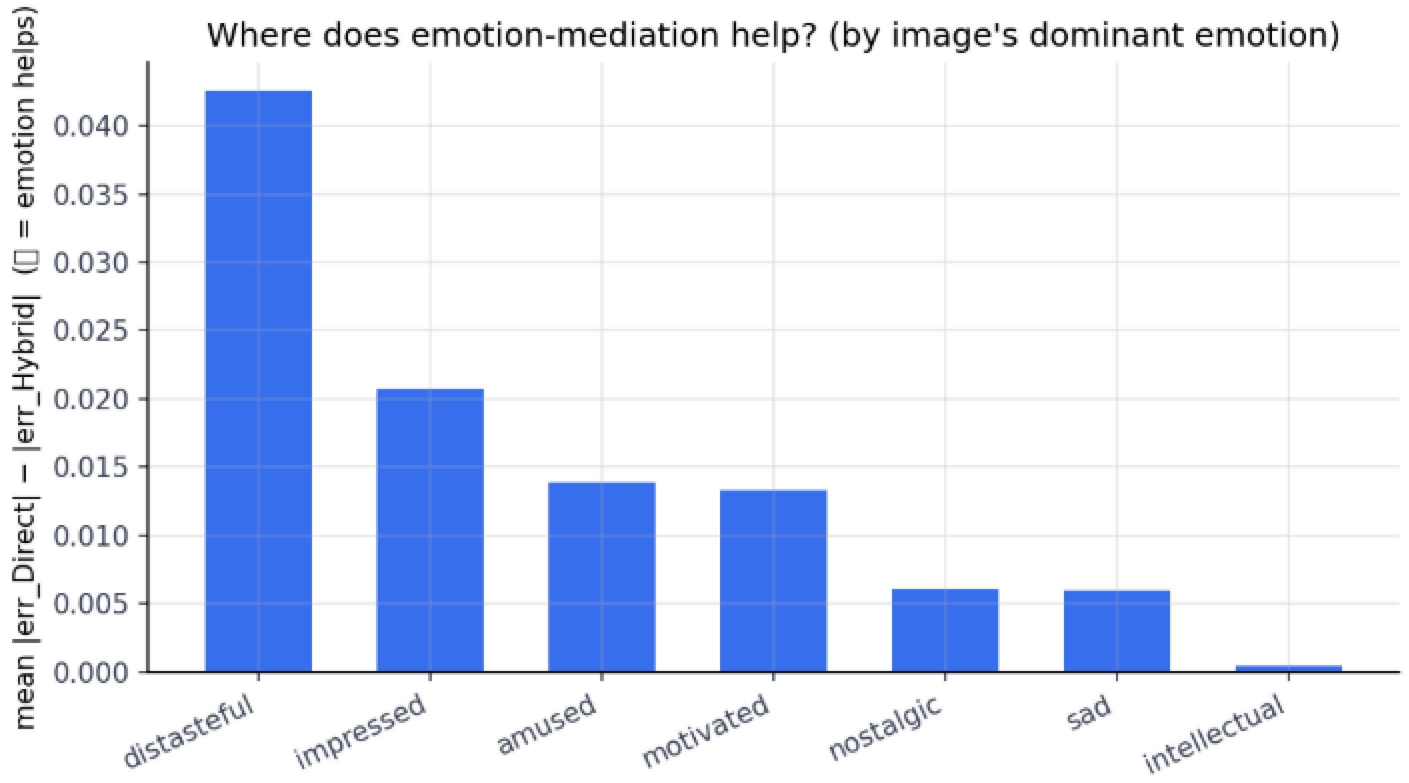
Distasteful	+0.043 (helps most)
Impressed	+0.021
Amused	+0.014
Nostalgic	+0.006
Intellectual	+0.001 (helps least)

STABILITY HYPOTHESIS & POWER LIMIT

Hypothesis: Images evoking stable/consistent emotions benefit more.

Spearman r = +0.39, n = 7	p = 0.38 (INCONCLUSIVE)

Honest Reporting: With only n=7 emotion categories, the statistical test lacks sufficient power. Direction matches hypothesis (+0.39), but result must be reported as inconclusive.



Unit of analysis changed from 129 users to 6,526 images categorized by dominant emotion label

Does the small model keep up?

MODEL VALIDATION & LAYER SWEEPING

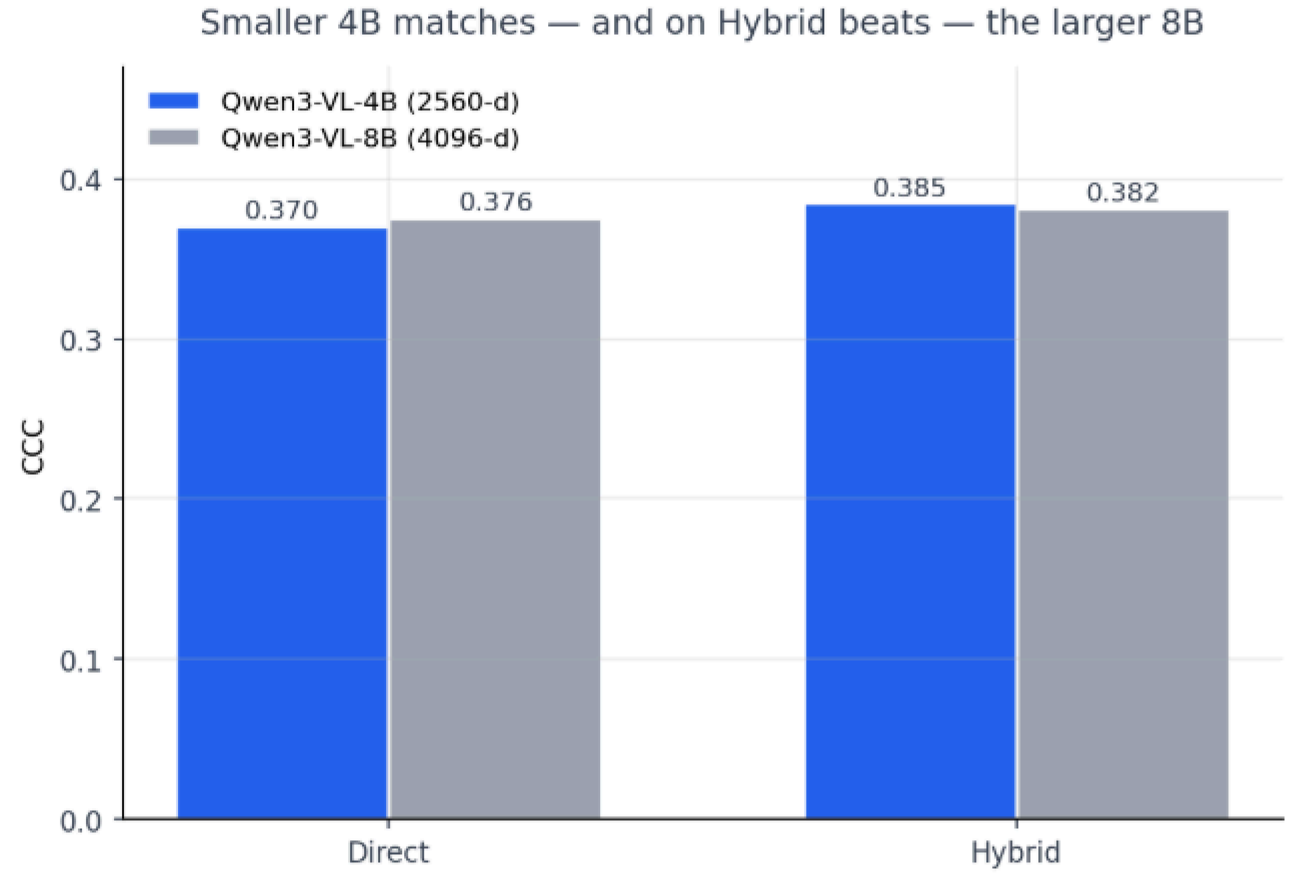
Ryu & Yanaka noted 4B performed best on AADB. We evaluated whether this holds on XPASS-Vis.

4B Best layer we found: LT17 | **8B Best layer we found:** LT15

PERFORMANCE COMPARISON (CCC)

Model (Dimensions)	Direct CCC	Hybrid CCC
Qwen4B (2560-d)	0.370	**0.385** ☆
Qwen8B (4096-d)	0.376	0.382

Efficiency Gain: The smaller 4B model (2,560 features) achieves slightly higher Hybrid CCC than 8B (4,096 features), saving significant GPU compute while maintaining performance.



LT = features extracted from text tokens · **number** = layer index · **sweep** = test all model layers per dataset

15 — PART 3 — THE PATTERN

The better the model, the less emotions add.

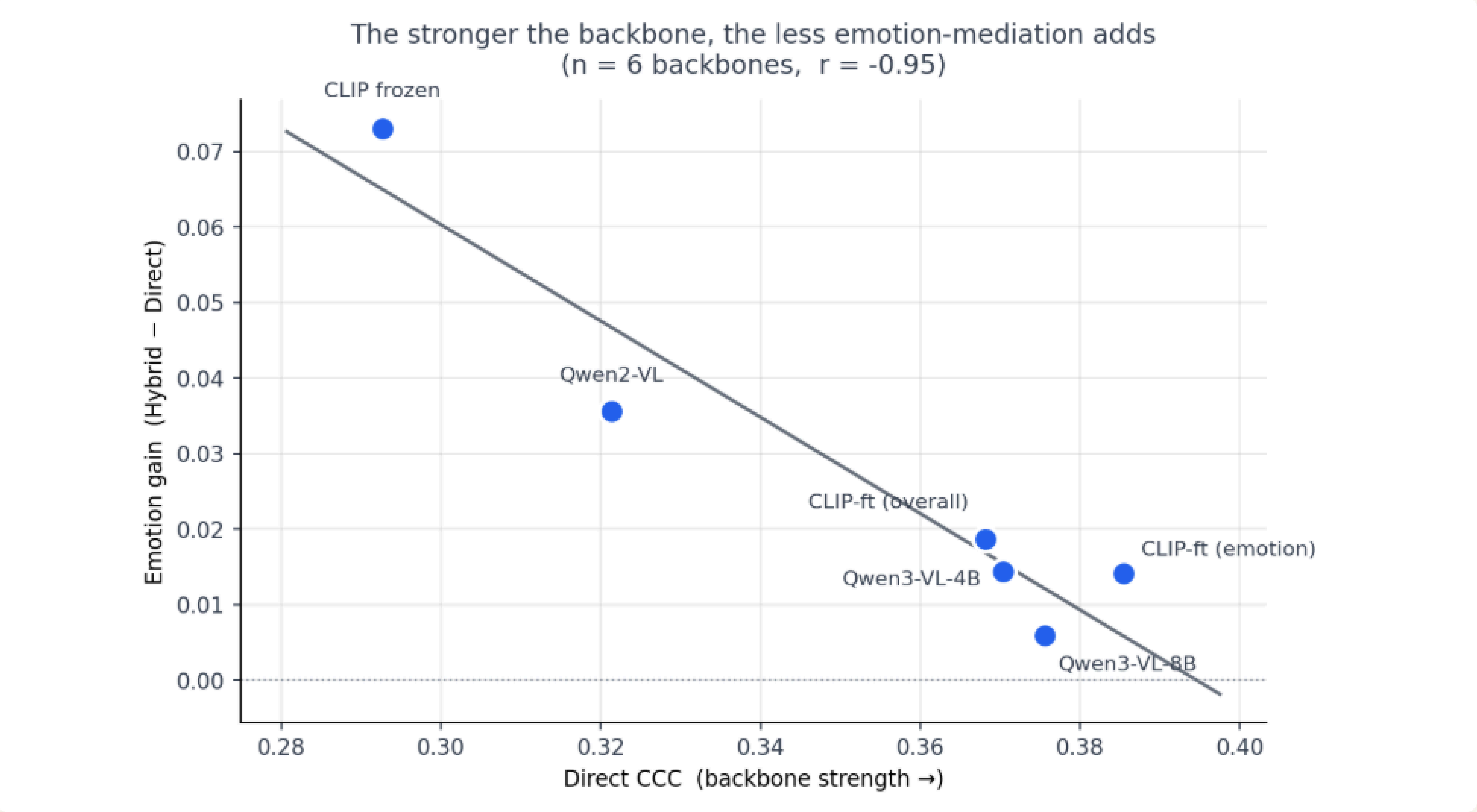
CORE FINDING ACROSS ALL TESTED BACKBONES

The stronger the vision backbone, the less explicit emotion-mediation adds on top.

Linear Correlation

r = −0.95

across 6 independent models



Implicit vs Explicit Representations

Strong VLMs already encode emotional semantics implicitly within their feature space. Making emotions explicit in Stage 2 yields diminishing returns as backbone capacity grows.

Multi-Backbone Rigor: This systematic trend only emerges when evaluating multiple diverse backbones (CLIP, ViT, ResNet, Qwen8B, Qwen4B) rather than relying on a single architecture.

6 backbones tested: CLIP-ViT-B/32, CLIP-ViT-L/14, ConvNeXt, Swin-v2, Qwen3-VL-8B, Qwen3-VL-4B

16 — SUMMARY

To sum up.

01 **The true human ceiling is about 0.69, not 0.725**

Measuring 4,509 image rating pairs where the same person rated the exact same photo twice shows a self-agreement of 0.693. The old 0.725 ceiling was inflated by shared mood noise because emotions and scores were rated in the same sitting.

02 **Emotion helps 92 percent of people**

Expected gain is a gradual effect driven by how well we read their feelings, not a fixed cutoff, and not about how strong the plain model is on its own. Even the hardest-to-read group still sees a small average gain (+0.031).

03 **Direct model strength is not the real driver of the gain**

Direct strength correlates with gains at $r = +0.15$, but it moves closely with emo_r at +0.66. Removing emo_r's influence via partial correlation drops Direct's correlation to −0.08 (non-significant), showing that only emotion accuracy truly matters.

04 **Art difference is a ceiling effect, not a new mechanism**

Art looks different mostly because it's already close to its ceiling, not because it works by a different mechanism—though the evidence there is borderline. Controlling for baseline strength via partial correlation restores significance (+0.175, $p=0.048$).

05 **Stronger backbones need emotion less**

A near-perfect negative correlation ($r = -0.95$) across 6 models confirms that powerful backbones already capture emotional features. The smaller 4B model holds up fine, adding a sixth data point to this clean pattern.

17 — NEXT STEPS

What comes next.

WHAT I PLAN TO TRY NEXT

1. **Cold-start analysis**
   How many ratings needed before personal model beats population formula?

2. **Trait-conditioned emotion prediction**
   Can trait vectors predict how a user's emotions deviate from population? If yes → zero-shot personalization.

Not in a hurry — can continue from Thailand after the internship if needed.

18 — DISCUSSION

What I would like your view on.

1. **Should we run cross-dataset validation?**
Should I validate the Hybrid pipeline on other datasets (like FLICKR-AES or PARA) to verify generalization, and do you have recommendations? Or is evaluating on a single unique dataset generally accepted in this community?

2. **Should we fine-tune Qwen-4B end-to-end?**
Is end-to-end fine-tuning on the recommended Qwen-4B model worth the computation for potential accuracy gains, or should we freeze the model here and start writing the paper?

3. **Does the ICI baseline actually benefit from traits?**
Should we run an ablation study by removing personality traits from the ICI baseline to verify whether its accuracy gain comes from traits or its architecture?

4. **What is our weakest point for an ACM MM submission?**
If we submit this work to ACM MM 2027, what do you think is our main weakness or the most likely reason reviewers might reject it?

5. **ACM MM vs. ACII: Which is more suitable?**
Which venue fits this project's profile better: a major multimedia conference like ACM MM, or an affective computing focus like ACII?

PROGRESS UPDATE #4

Thank You

Pinwa · JAIST Internship

Backup: how the analysis is set up.

QA-1 How is delta computed?

A per-user pairwise difference, taken for every user in every domain from results that already existed.

$$\Delta_{u,d} = \mathrm{CCC}_{\mathrm{Hybrid}\,u,d} - \mathrm{CCC}_{\mathrm{Direct}\,u,d}$$

Data source	Rows	Retraining
exp1_per_user	**387** 129 users × 3 domains	**Zero** secondary analysis

Strict controls: Hybrid and Direct come from the exact same test split, the same user ratings and identical folds.

QA-2 Isn't Direct CCC a second cause?

A backbone that represents images well is good at **both** direct scoring and emotion prediction, so the two move together (r = +0.66).

Uncontrolled correlation	Partial, controlling emo_r
+0.15 looks like a second factor	**−0.04** effect vanishes

Classic confound: Like shoe size correlating with reading ability when age is the real cause. Direct CCC tracks delta only because both are driven by emo_r.

QA-3 Is there any data leakage?

The pipeline is fitted in two stages, on two groups of users that never overlap.

Stage 1 · General set, 104 users
Trained exclusively on general users to learn the generic image → emotion mapping.

Stage 2 · Target set, 25 users
Personal ridge model fit strictly per target user, on their own training fold only.

Zero user overlap: Leakage would require seeing target-user ratings during Stage 1, which this split makes impossible.

QA-4 What exactly is emo_r?

Emotion predictability for one specific user: how well Stage 1 recovers their 7 emotion ratings.

$$\mathrm{emo\_r}(u) = \frac{1}{7}\sum_{e=1}^{7} r(\hat{e}_e, e_e)$$

Stage 1 bottleneck	Stage 2 headroom
0.27 mean emotion accuracy	**0.693** P-oracle, true emotions

The bottleneck: Stage 2 could reach 0.69 given perfect emotions, so the pipeline is capped by Stage 1.

Backup: statistics and metrics.

QA-5 p = 0.056 — proven or not?

$\Delta \sim \text{emo\_r} \times \text{domain} \rightarrow p = 0.056$ (not significant)

Formal decision
We fail to reject the null at alpha = 0.05. We do **not** have statistical evidence that the mechanism in art differs from fashion or landscape.

Scientific nuance
Failing to reject is **not** proof the mechanisms are identical. It sits on the border, so we state it as inconclusive, favouring ceiling.

Methodological rigor: Absence of evidence is not evidence of absence. We report the value transparently.

QA-6 Did you correct for multiple comparisons?

28 candidate predictors of delta were screened, so the significance threshold has to be tightened.

$$\alpha_{adj} = \frac{0.05}{28} = 0.00179$$

Predictors	Method	Survivor
28	**Bonferroni**	**emo_r**
traits, stats, baselines		p = 1e-9

Robust significance: Only emotion-reading accuracy survives strict correction, accounting for $R^2 = 0.13$ of delta variance.

QA-7 Which cases failed?

Characterising the 32 user × domain units where emotion mediation made the prediction worse.

Failure proportion	Mean emo_r, hurt	Mean emo_r, helped
32 / 387	**0.153**	**0.283**
8.3% of units	read badly	read well

Root cause: Mediation fails almost exclusively where Stage 1 misses the user's emotion profile, which injects noise into Stage 2.

QA-8 Why CCC and not RMSE or Spearman?

Three candidate metrics, and what each one is blind to.

Lin's CCC
Ranking agreement **and** absolute scale. Punishes the right order at the wrong scale.

RMSE / MSE
Absolute error only. Does not isolate linear agreement or ranking quality.

Spearman rank
Ordinal ranking only, ignores scale entirely. Reported as a secondary metric.

Literature standard: CCC is the benchmark used by Hayashi-san et al. (XPASS-Vis) and Ryu & Yanaka (2024).